



Estimate with FIMS : : CHEATSHEET

GET FIMS
`install.packages("FIMS", repos = c("https://noaa-fisheries-integrated-toolbox.r-universe.dev", "https://cloud.r-project.org"))`

TIDY THE INPUT FIMSFrame

Prepare one tidy data frame for your FIMS model

```
model_inputs <-  
FIMSFrame(your_long_data_frame)
```

Investigate the FIMSFrame class
`get_*`() functions extract information from a FIMSFrame object

```
data("data_big", package = "FIMS")  
data_4_model <- FIMSFrame(data_big)  
get_n_fleets(data_4_model)  
[1] 2  
get_ages(data_4_model)  
[1] 1 2 3 4 5 6 7 8 9 10 11 12  
get_data(data_4_model)
```

Package data

Structure of the long, data format

type	fleet	timing	observed	unit	uncertainty	age
landings	trawl	2025	1000	mt	"~dlnorm(meanlog = log_landings_expected, sdlog = 0.0001)"	NA
landings	other_fleet	2024	2366	mt	"~dlnorm(meanlog = log_landings_expected, sdlog = 0.0001)"	NA
age_comp	trawl	2025	0.1	proportion	"~dmultinom(prob = agecomp_proportion, size = 100)"	1
age_comp	trawl	2025	0.2	proportion	"~dmultinom(prob = agecomp_proportion, size = 100)"	2
...	...	...	...	...	...	...

type: landings, index, age_comp, length_comp, weight_at_age, age_to_length_conversion
fleet: name of the fleet, can be any string
timing: double values specifying the collection timing
observed: value for the measurement
unit: unit of the measurement
uncertainty: formula that specifies the distribution of the observation and parameters for that distribution
age: desired binning structures for your model (the data frame can have additional binning columns such as **length** and, in the future, **sex**)

FIT A MODEL FIMSFit

Default parameters based on data

```
fit <- setup_default_parameters(data = model_inputs) |>  
  initialize_fims(  
    parameters = parameters,  
    data = model_inputs  
  ) |>  
  fit_fims(optimize = TRUE)
```

Send all of the specified parameters and data to their respective C++ modules

Optimize the model using nlminb as the default

Default model configurations are created based on the data, e.g., logistic selectivity for each fleet in your data

See [vignette](#) for details on modifying the parameters tibble

Investigate the FIMSFit class `get_*`() functions extracts information from a FIMSFit object

```
get_estimates(fit)  
get_number_of_parameters(fit)  
get_run_time(fit)
```

```
# TMB-specific information  
get_report(fit)  
get_obj(fit)  
get_par(fit)
```

